

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering
Academic Year 2025-26

Rubrics for Lab Experiments

Class : *B.E. Computer Engineering*

Subject Name : *BDA Lab*

Semester : VII

Subject Code : *CSL702*

Practical No:	3
Title:	Write a program in python to implement Word Count using MapReduce
Date of Performance:	23-7-25
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation:

Performance Indicator	Below average	Average	Good	Excellent	Marks
On time Submission (2)	Not submitted (0)	Submitted after deadline (1)	Early or on time submission(2)	---	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

Aim: Write a program in python to implement Word Count using MapReduce

Objective:

- Understand WordCountAggregationAlgorithm
- Use MapReduce framework for WordCount
- Write python program to find WordCount

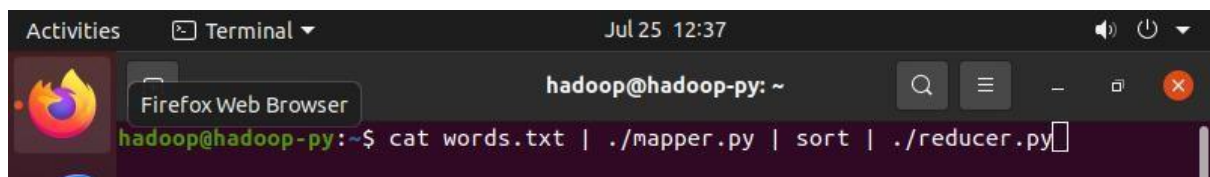
Input:

Subject name is Big data analytics. Big data analytics Subject is in final year.

Output:

Subject 2
name 1
is 2
Big 2
data 2
analytics 2
in 1
final 1
year 1

Result: Implementation of the solution for above problem and attach input, program, intermediate and output files.

A screenshot of a Linux terminal window. The window title is "Terminal" and the date/time is "Jul 25 12:37". The prompt is "hadoop@hadoop-py: ~". The command entered is "cat words.txt | ./mapper.py | sort | ./reducer.py". The command is partially executed, with a cursor at the end. The terminal has a dark background and a light-colored cursor.

```
hadoop@hadoop-py: ~$ cat words.txt | ./mapper.py | sort | ./reducer.py
```

```
Activities  Terminal Jul 25 12:38
hadoop@hadoop-py: ~
statement. 1
Statistics 2
stay 2
step 1
Stephen 1
steps 1
stick 1
stops 1
stressed 1
stronger 1
structuring 1
struggling, 1
student 4
students 3
studied 1
subject. 1
subjects 1
succeed 2
success 1
successful 3
successful. 1
successfully. 1
such 4
Such 2
supervise 1
support 1
system 1
systems 1
take 1
```

```
Activities  Terminal Jul 25 12:38
hadoop@hadoop-py: ~
with, 1
within 1
without 3
won't 1
word 8
words 2
words. 2
words? 2
work. 2
working 2
Working 1
worldwide 1
write 3
writers 2
writing 3
writing, 2
writing. 1
Writing 1
Writing? 1
www.mindtools.com/pages/article/newHTE_00.htm. 1
www.nilc.co.uk/top-10-soft-skills-that-employers-will-be-looking-for-in-2022 1
XXI 1
you 10
You 3
your 4
you're 1
yourself: 2
you've 1
hadoop@hadoop-py:~$
```

Mapper.py

```
import sys
for line in sys.stdin:
    # to remove leading and trailing whitespace
    line = line.strip()
    # split the line into words
    words = line.split()

    for word in words:
        print('%s\t%s' % (word, 1))
```

Reducer.py

```
from operator import itemgetter
import sys

current_word = None
current_count = 0
word = None

for line in sys.stdin:
    line = line.strip()
    word, count = line.split('\t', 1)
    try:
        count = int(count)
    except ValueError:
        continue

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print('%s\t%s' % (current_word, current_count))
            current_count = count

        current_word = word

    if current_word == word:
        print('%s\t%s' % (current_word, current_count))
```